

VIRGINIA COMMONWEALTH UNIVERSITY

ASSIGNMENT REPORT

Exploratory Data Analysis

Course Code: MBA V 443M

Course Title: Statistical Modelling and AI-Driven Analytics

Assignment No: Assignment 1

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1. Introduction

The Student Dropout Dataset contains demographic, academic, and socio-economic information about students, along with their final status (Dropout, Graduate, or Enrolled). The purpose of this assignment is to perform Exploratory Data Analysis (EDA) using Python to uncover patterns, assess data quality, and identify predictors of dropout. Insights from this analysis can help educational institutions design targeted interventions to improve retention and student success.

2. Data Understanding and Quality Assessment

The dataset includes variables such as marital status, course enrollment, attendance type, prior qualifications, parental education and occupation, admission grades, tuition fee status, scholarship, age, and macroeconomic indicators (unemployment rate, inflation, GDP). Data quality checks (`info()`, `describe()`, `isnull()`, `drop_duplicates()`) confirmed:

- No missing values in key columns.
- Delimiter issue (semicolon) was corrected during loading.
- The target variable contained three categories: Dropout, Graduate, Enrolled.
- Column names required cleaning (spaces and capitalization).

3. Methodology

The analysis was conducted in Jupyter Notebook using Python libraries:

- Pandas, NumPy for data handling
- Matplotlib, Seaborn for visualization

Steps followed:

1. Loading and inspecting the dataset with correct delimiter.
2. Cleaning column names and encoding categorical variables.
3. Descriptive statistics to summarize numerical variables.
4. Visualizing distributions (histograms, KDE, bar charts).
5. Exploring categorical variables (marital status, tuition fee status).
6. Correlation analysis among academic and economic indicators.
7. Building predictive models: Logistic Regression as baseline, with exploratory trials of Decision Trees and Random Forests for feature importance.
8. Evaluating model performance using classification metrics.

4. Visualization

Several visualization techniques were applied:

- **KDE Plot of Inflation Rate** → showed multimodal distribution, highlighting variability in macroeconomic conditions.
- **Bar Chart of Student Outcomes by Marital Status** → revealed that most students fall under marital status code 1, with graduates dominating, followed by dropouts and enrolled.
- **Histogram of Unemployment Rate** → indicated common unemployment ranges around 10–12%, providing socio-economic context.
- **Target Distribution Plot** → showed graduates as the largest group, followed by dropouts and enrolled students.
- **Correlation Heatmap** → highlighted strong relationships between curricular unit performance and dropout status.

Key Insights

- Curricular unit grades and tuition fee status emerged as the strongest predictors of dropout.
- Marital status code 1 students formed the largest group, with graduates dominating, but dropout risk was still significant.
- Macroeconomic indicators (unemployment, inflation) provided background context but were weaker predictors compared to academic performance.
- Correlation analysis showed strong links between admission grades, curricular unit performance, and dropout likelihood.
- Outliers in grades and tuition fee status suggested specific high-risk student segments.

Conclusions

The EDA revealed that dropout is influenced primarily by academic performance and financial status, with socio-economic and demographic factors providing secondary context. Institutions should prioritize:

- Academic mentoring for low-performing students.
- Financial aid programs for students with unpaid tuition.
- Targeted support for high-risk groups identified through categorical analysis.

This analysis demonstrates how EDA transforms raw student data into actionable insights for retention strategies. Future work could extend to predictive modeling using ensemble methods for improved accuracy.